

Для последовательности значений $x_n = 100 \cos n, n = \overline{1, N}$ реализовать преобразование $\{x_n\}_{n=1}^N \rightarrow \{y_n\}_{n=1}^N, y_n = f(x_n)$. Принять $N = 2000$. Сравнить эффективность выполнения данного преобразования на основе шаблонов `std::transform` и `concurrency::parallel_transform`. В качестве контейнеров для хранения последовательностей использовать: а) вектор `std::vector`, б) дек `std::deque`, в) список `std::list`. Полагать:

1.

$$f(x) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2}{x^2 + k^2 + j^2} \cos((k+j)x) \approx$$

$$\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2}{x^2 + k^2 + j^2} \cos((k+j)x)$$

2.

$$f(x) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2}{x^2 + 2k^2 + j^2} \sin((k+2j)x) \approx$$

$$\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2}{x^2 + 2k^2 + j^2} \sin((k+2j)x)$$

3.

$$f(x) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2}{x^2 + k^2 + 2j^2} \cos kx \sin jx \approx$$

$$\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2}{x^2 + k^2 + 2j^2} \cos kx \sin jx$$

4.

$$f(x) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2 k}{x^2 + k^3 + j^2} \cos((k+j)x) \approx$$

$$\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2 k}{x^2 + k^3 + j^2} \cos((k+j)x)$$

5.

$$f(x) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2 j}{x^2 + k^2 + j^3} \sin((k+2j)x) \approx$$

$$\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2 j}{x^2 + k^2 + j^3} \sin((k+2j)x)$$

6.

$$f(x) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2 (k+j)}{x^2 + k^3 + j^3} \cos kx \sin jx \approx$$

$$\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2 (k+j)}{x^2 + k^3 + j^3} \cos kx \sin jx$$

7.

$$\begin{aligned}
f(x) &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{(x^2 + x)(k - j)}{x^2 + k^3 + j^3} \sin kx \cos jx \approx \\
&\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{(x^2 + x)(k - j)}{x^2 + k^3 + j^3} \sin kx \cos jx
\end{aligned}$$

8.

$$\begin{aligned}
f(x) &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{(x^2 - x)(k + 2j)}{x^2 + k^3 + j^3} \cos((k + 2j)x) \approx \\
&\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{(x^2 - x)(k + 2j)}{x^2 + k^3 + j^3} \cos((k + 2j)x)
\end{aligned}$$

9.

$$\begin{aligned}
f(x) &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2(k^2 + j^2)}{x^2 + k^4 + j^4} \cos((k - j)x) \approx \\
&\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2(k^2 + j^2)}{x^2 + k^4 + j^4} \cos((k - j)x)
\end{aligned}$$

10.

$$\begin{aligned}
f(x) &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2(k^2 + j)}{x^2 + k^4 + j^3} \sin((2k - j)x) \approx \\
&\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2(k^2 + j)}{x^2 + k^4 + j^3} \sin((2k - j)x)
\end{aligned}$$

11.

$$\begin{aligned}
f(x) &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \frac{x^2(k + 2j^2)}{x^2 + k^3 + j^4} \cos((k + 2j)x) \approx \\
&\approx \sum_{k=1}^{\max(20, [20|x|])} \sum_{j=1}^{\max(20, [20|x|])} \frac{x^2(k + 2j^2)}{x^2 + k^3 + j^4} \cos((k + 2j)x)
\end{aligned}$$